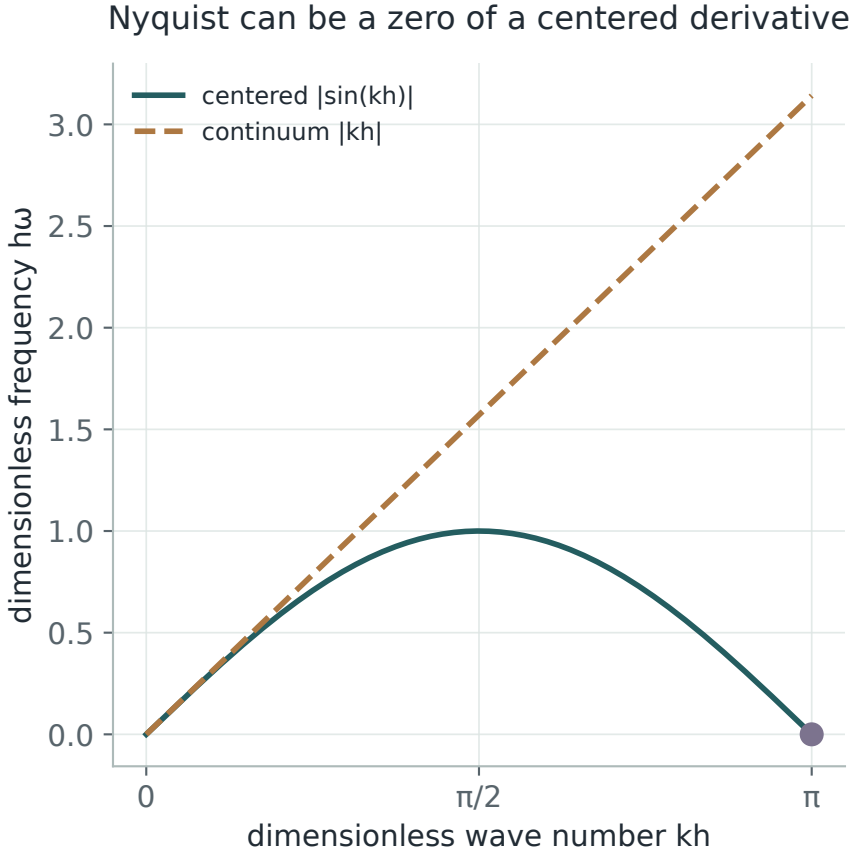
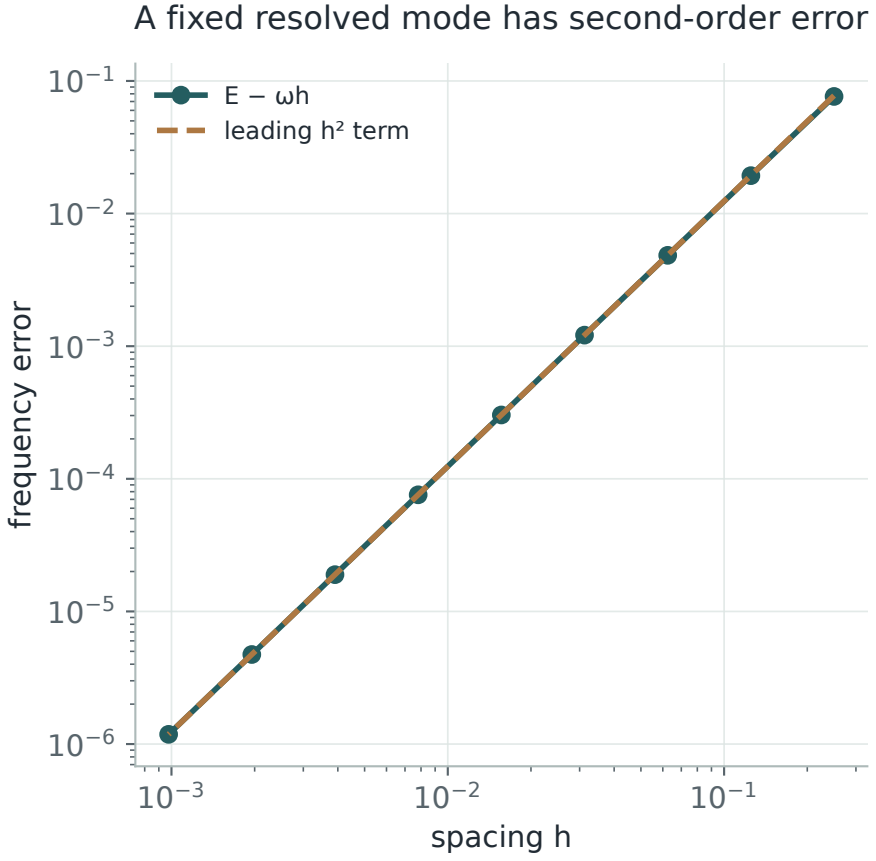

Nyquist can be a zero of a centered derivative
centered |sin(kh)|
continuum |kh|
dimensionless frequency hω
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
π/2
π
dimensionless wave number kh


A fixed resolved mode has second-order error
E − ωh
leading h² term
frequency error
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
10⁻⁶
10⁻³
10⁻²
10⁻¹
spacing h